

Assume the following throughout this entire sheet

- $\mathbf{F}(x, y, z) = [M(x, y, z), N(x, y, z), P(x, y, z)]$ a vector-valued function in $\mathbf{R}^3$ with continuous partial derivatives.
- $\mathbf{G}(x, y) = [M(x, y), N(x, y)]$ a vector-valued function in $\mathbf{R}^2$ with continuous partial derivatives.
- $g(x, y, z)$ a scalar-valued function with continuous first partial derivatives
- $\mathbf{r}(t)$ represents some parameterization of a curve C for $a \leq t \leq b$
- C_0 is a simple closed curve enclosing a region R

Operators: $\nabla g(x, y, z) = [g_x, g_y, g_z]$ $\operatorname{div} \mathbf{F} = M_x + N_y + P_z$ $\operatorname{curl} \mathbf{F} = [P_y - N_z, M_z - P_x, N_x - M_y]$

Line Integrals: Let $\mathbf{r}(t)$ be a parameterization for the curve C

$$\text{Vector-Valued: } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \quad \text{Scalar-Valued: } \int_C g ds = \int_a^b g(\mathbf{r}(t)) \cdot |\mathbf{r}'(t)| dt$$

Surface Integrals: Let $\mathbf{r}(u, v)$ be a parameterization for the surface S

$$\text{Vector-Valued: } \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R \mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} du dv \quad \text{Scalar-Valued: } \iint_R g(\mathbf{r}(u, v)) \cdot |\mathbf{N}| du dv$$

Stokes' Theorem: Let C be a closed curve which is the boundary for a surface S . Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS$$

* Recall: Green's Theorem is a special case of Stokes' Theorem where S is in the xy -plane and $\mathbf{n} = [0, 0, 1]$.

Divergence Theorem: Let S be a surface that is the boundary of T . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_T \operatorname{div} \mathbf{F} dV$$

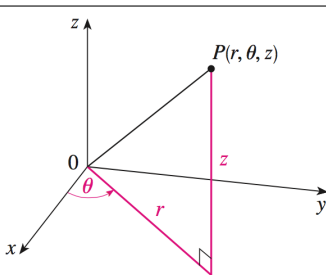
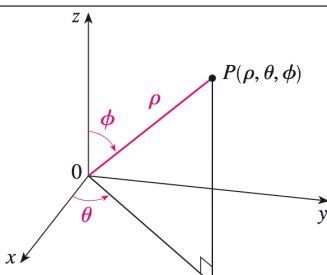
Complex Numbers:

Let $z = re^{\theta i} = r(\cos \theta + i \sin \theta)$

$$z^n = r^n e^{n\theta i} = r^n [\cos(n\theta) + i \sin(n\theta)]$$

$$z^{1/n} = r^{1/n} e^{\frac{\theta + 2\pi k}{n} i} = r^{1/n} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right] \quad \text{for } k = 0, 1, 2, \dots, n-1$$

Special Coordinates

Cylindrical		Spherical	
$x = r \cos \theta$ $y = r \sin \theta$ $z = z$		$x = \rho \sin \phi \cos \theta$ $y = \rho \sin \phi \sin \theta$ $z = \rho \cos \phi$	
$x^2 + y^2 = r^2$ $dV = r dr d\theta dz$		$x^2 + y^2 + z^2 = \rho^2$ $dV = \rho^2 \sin \phi d\rho d\theta d\phi$	